

Estimating Differences and Checking Your Answer

Grades 2–3 Number and Operations in Base Ten

Estimating differences by rounding, judging whether an answer is reasonable, and checking a difference by adding

Three tools you will use today.

- **Estimate.** Round *each* number to the same place, then subtract the rounded numbers. Use $\approx$ (“is about”) instead of $=$.
- **Is it reasonable?** An exact answer should land close to the estimate. If it is far away, a trade was probably missed.
- **Check by adding.** Addition undoes subtraction. If $\text{start} - \text{taken away} = \text{answer}$, then $\text{answer} + \text{taken away}$ must give back the start.

Worked model: A box holds 506 crayons. 291 crayons are taken out. About how many are left?

Round to the nearest hundred: $506 \approx 500$ and $291 \approx 300$, so $500 - 300 = 200$ crayons, about 200 crayons are left. The exact answer is 215 crayons. Because 215 is close to 200, the answer is reasonable. Check it: $215 + 291 = 506$ crayons.

Directions. Read each story carefully. Show your rounding or your trades in the work space. Write the number the question asks for on the answer line, and label your answer with what it counts.

1. The school library has 412 books on the shelf. Students check out 187 books. Round *each* number to the nearest hundred, then subtract the rounded numbers to estimate how many books are still on the shelf.

$$412 - 187 \approx \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

Answer: _____

2. A ferry can carry 265 passengers. On Monday 138 passengers rode it. Round *each* number to the nearest ten, then subtract the rounded numbers to estimate how many more passengers the ferry could have carried.

$$265 - 138 \approx \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

Answer:

3. Mr. Diaz started with a crate of apples. He sold 165 apples and says 175 apples are left. Check his subtraction by adding his answer back to the apples he sold: find $175 + 165$. Write the sum on the answer line, then write one sentence saying how many apples were in the crate at the start.

Answer:

4. A stadium has 738 seats. During a game 264 seats are empty. Round *each* number to the nearest hundred, then subtract to estimate how many seats are filled.

$$738 - 264 \approx \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

Answer:

5. A recycling club wants to collect 906 cans. So far the club has collected 497 cans.

- (a) Estimate how many more cans are needed by rounding each number to the nearest hundred.
- (b) Now find the *exact* number of cans still needed.
- (c) Is your exact answer close to your estimate? Write yes or no.

Write the estimate and then the exact number on the answer line.

Answer: _____

6. A book has 623 pages. Ben has read 175 pages. Ben says 552 pages are left.

- (a) Check Ben's answer by adding: find $552 + 175$.
- (b) Does the check give back 623 pages? If it does not, find the exact number of pages left.

Write the sum and then the exact number of pages on the answer line.

Answer: _____

- $$851 - 349 \approx \underline{\hspace{2cm}} - \underline{\hspace{2cm}}$$

Answer: _____

- Estimate the number of adult tickets by rounding each number to the nearest hundred.
- Find the exact number of adult tickets.

Answer: _____

9. Rico had 703 stickers and gave away 158 stickers. He wrote that 655 stickers were left. Estimate the difference to the nearest hundred and use your estimate to explain, in one sentence, why 655 stickers cannot be right. Then work $703 - 158$ correctly and write the exact number of stickers on the answer line.

Answer: _____

10. A food drive has a goal of 900 cans. Volunteers collected 348 cans in week one and 275 cans in week two. Round every number to the nearest hundred, then estimate how many more cans are still needed to reach the goal.

$$900 - 348 - 275 \approx \underline{\hspace{1cm}} - \underline{\hspace{1cm}} - \underline{\hspace{1cm}}$$

Answer: _____